Q1 (20 July 2021 Shift 1)

A radioactive material decays by simultaneous emissions of two particles with half lives of 1400 years and 700 years respectively. What will be the time after the which one third of the material remains ? (Take $\ln 3 = 1.1$)

(1) 1110 years

(2) 700 years

(3) 340 years

(4) 740 years

Q2 (20 July 2021 Shift 1)

A nucleus of mass M emits γ -ray photon of frequency ' v'. The loss of internal energy by the nucleus is: [Take 'c' as the speed of electromagnetic wave]

(1) hv

(2) 0

(3) hv $\left[1-\frac{hv}{2Mc^2}\right]$

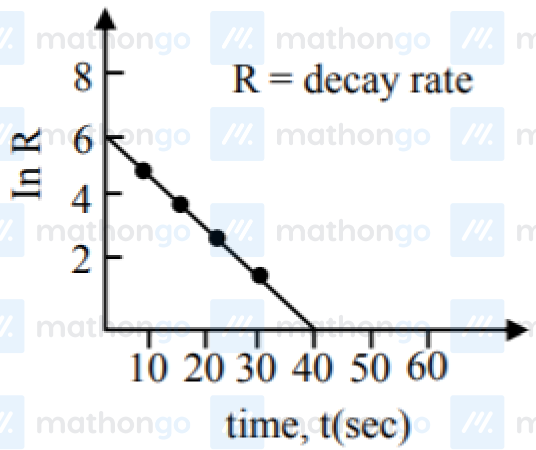
(4) hv $\left[1+\frac{hv}{2Mc^2}\right]$

Q3 (20 July 2021 Shift 2)

For a certain radioactive process the graph between

In R and t(sec) is obtained as shown in the figure.

Then the value of half life for the unknown radioactive material is approximately :



(1) 9.15sec

(2) 6.93sec

(3) 2.62sec

(4) 4.62sec

Q4 (20 July 2021 Shift 2)

A radioactive substance decays to $\left(\frac{1}{16}\right)^{\text{th}}$ of its

initial activity in 80 days. The half life of the radioactive substance expressed in days is ____.

Q5 (22 July 2021 Shift 1)

A nucleus with mass number 184 initially at rest emits an α -particle. If the Q value of the reaction is 5.5MeV, calculate the kinetic energy of the $\alpha-$ particle.

(1) 5.0MeV

(2) 5.5MeV

(3) 0.12MeV

(4) 5.38MeV

Q6 (25 July 2021 Shift 1)

Some nuclei of a radioactive material are undergoing radioactive decay. The time gap between the instances when a quarter of the nuclei have decayed and when half of the nuclei have decayed is given as:

(where λ is the decay constant)

(1) $\frac{1}{2} \frac{\ln 2}{\lambda}$

(2) $\frac{\ln 2}{\lambda}$

(3) $\frac{2 l n 2}{\lambda}$

(4) $\frac{\ln \frac{\mathrm{d}}{2}}{2}$

Q7 (25 July 2021 Shift 1)

The half-life of ${}^{198}\mathrm{Au}$ is 3 days. If atomic weight of ${}^{198}\mathrm{Au}$ is 198 g/mol then the activity of 2mg of ${}^{198}\mathrm{Au}$ is [in disintegration/second] :

(1) 2.67×10^{12}

(2) 6.06×10^{18}

(3) 32.36×10^{12}

(4) 16.18×10^{12}

Q8 (25 July 2021 Shift 2)

From the given data, the amount of energy required to break the nucleus of aluminium ${}_{13}^{27}\mathrm{Al}$ is $\mathrm{x} \times 10^{-3}$ J

Mass of neutron $= 1.00866\mathrm{u}$

Mass of proton $= 1.00726\mathrm{u}$

Mass of Aluminium nucleus $= 27.18846\mathrm{u}$

(Assume 1 u corresponds to x J of energy) (Round off to the nearest integer)

Q9 (25 July 2021 Shift 2)

The nuclear activity of a radioactive element becomes $\left(\frac{1}{8}\right)^{\text{th}}$ of its initial value in 30 years. The half-life of radioactive element is ___ years.

Q10 (27 July 2021 Shift 1)

If 'f' denotes the ratio of the number of nuclei decayed $(\mathrm{N_d})$ to the number of nuclei at $\mathrm{t}=0$ $(\mathrm{N_0})$ then for a collection of radioactive nuclei, the rate of change of ' f with respect to time is given as:

[λ is the radioactive decay constant]

(1) $-\lambda\left(1-\mathrm{e}^{-\lambda t}\right)$

(2) $\lambda\left(1-\mathrm{e}^{-\lambda t}\right)$

(3) $\lambda \mathrm{e}^{-\lambda t}$

(4) $-\lambda \mathrm{e}^{-\lambda \mathrm{d}}$

Q11 (27 July 2021 Shift 1)

A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance $200 \mu\mathrm{F}$ is first charged and later connected with resistor $'\mathrm{R}'$. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be ___ Ω.

Q12 (27 July 2021 Shift 2)

Consider the following statements:

A. Atoms of each element emit characteristics

spectrum.

B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C. The density of nuclear matter depends on the size of the nucleus.

D. A free neutron is stable but a free proton decay is possible.

E. Radioactivity is an indication of the instability of nuclei. Choose the correct answer from the options given below :

(1) A, B, C, D and E

(2) A, B and E only

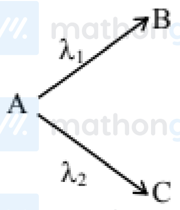
(3) B and D only

(4) A, C and E only

Answer Key

Q1 (4)	**Q2** (4)	**Q3** (4)	**Q4** (20)
Q5 (4)	**Q6** (4)	**Q7** (4)	**Q8** (27)
Q9 (10)	**Q10** (3)	**Q11** (150)	**Q12** (2)

Q1



Given $\lambda_1 = \frac{\ln 2}{700}$ / year, $\lambda_2 = \frac{\ln 2}{1400}$ / year

$\therefore \lambda_{\text{met}} = \lambda_1 + \lambda_2 = \ell n2\left[\frac{1}{700} + \frac{1}{1400}\right]$

$= \frac{3\ell n2}{1400}$ / year

Now, Let initial no. of radioactive nuclei be

No.

$\therefore \frac{N_0}{3} = N_0 e^{-\lambda_{\text{mat}}}$

$\Rightarrow \ell n\frac{1}{3} = -\lambda_{\text{net}} t$

$\Rightarrow 1.1 = \frac{3 \times 0.693}{1400} t \Rightarrow t \approx 740$ years

Hence option 4

Q2

Energy of γ ray $[E_\gamma] = hv$ Momentum of γ ray $[P_\gamma] = \frac{h}{\lambda} = \frac{hv}{C}$

Total momentum is conserved. $\overrightarrow{P}_y + \overrightarrow{P}_{Nu} = 0$

Where $\overrightarrow{P}_{Nu}$ = Momentum of decayed nuclei $\Rightarrow P_\gamma = P_{Nu}$

$\Rightarrow \frac{hv}{C} = P_{Nu}$

$\Rightarrow$ K. E. of nuclei

$= \frac{1}{2}Mv^2 = \frac{(P_{Nu})^2}{2M} = \frac{1}{2M}\left[\frac{hv}{C}\right]^2$

Loss in internal energy $= E_\gamma + K \cdot E_{Nu}$

$= hv + \frac{1}{2M}\left[\frac{hv}{C}\right]^2$

$= hv\left[1 + \frac{hv}{2MC^2}\right]$

Q3

$R = R_0 e^{-\lambda t}$

$\ell n R = \ell n R_0 - \lambda t$

$-\lambda$ is slope of straight line

$\lambda = \frac{3}{20}$

$t_{1/2} = \frac{\ln 2}{\lambda} = 4.62$

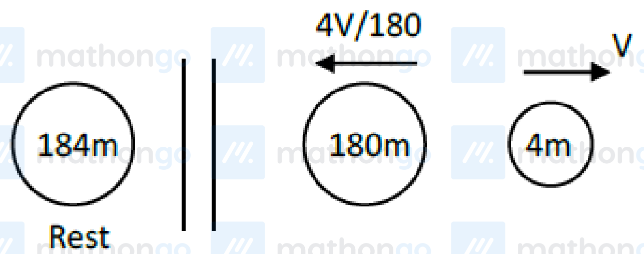
Q4

$$N_0 \xrightarrow{t_{\frac{1}{2}}} \frac{N_0}{2} \xrightarrow{t_{\frac{1}{2}}} \frac{N_0}{4} \xrightarrow{t_{\frac{1}{2}}} \frac{N_0}{8} \xrightarrow{t_{\frac{1}{2}}} \frac{N_0}{16}$$

$4 \times t_{1/2} = 80$

$t_{1/2} = 20$ days

Q5



$$\frac{1}{2}(4m)v^2 + \frac{1}{2}(180m)\left(\frac{4v}{180}\right)^2 = 5.5\text{MeV}$$

$$\Rightarrow \frac{1}{2}4mv^2\left[1 + 45\left(\frac{4}{180}\right)^2\right] = 5.5\text{MeV}$$

$$\Rightarrow K.E_\alpha = \frac{5.5}{1 + 45\cdot\left(\frac{4}{180}\right)^2}\text{MeV}$$

$$K.E_\alpha = 5.38\text{MeV}$$

Q6

$\frac{3\,\mathrm{N}_0}{4}=\mathrm{N}_0\mathrm{e}^{-it_1}$

$\frac{\mathrm{N}_0}{2}=\mathrm{N}_0\mathrm{e}^{-\lambda_2}$

$\ln(3/4)=-\lambda\mathrm{t}_1$

$\ln(1/2)=-\lambda t_2 \quad \ldots.(i)$

$\ln(3/4)-\ln(1/2)=\lambda\left(\mathrm{t}_2-t_1\right)$

$\Delta t=\frac{\ln(3/2)}{\lambda}$

Q7

$A=\lambda N$

$\lambda=\frac{\ln 2}{t_{1/2}}=\frac{\ln 2}{3\times 24\times 60\times 60}\sec^{-1}=2.67\times 10^{-6}\sec^{-1}$

N = Number of atoms in 2mg Au

$=\frac{2\times 10^{-3}}{198}\times 6\times 10^{23}=6.06\times 10^{15}$

$\mathrm{A}=\lambda\mathrm{N}=1.618\times 10^{13}=16.18\times 10^{12}\mathrm{dps}$

Q8

$\Delta m=\left(Zm_P+(A-Z)m_n\right)-M_{Me}$

$=(13\times 1.00726+14\times 1.00866)-27.18846$

$=27.21562-27.18846$

$=0.02716\mathrm{u}$

$\mathrm{E}=27.16\mathrm{x}\times 10^{-3}\ \mathrm{J}$

Q9

$\mathrm{A}=\mathrm{A}_0\mathrm{e}^{-\lambda.t}$

$\frac{\mathrm{A}_0}{8}=\mathrm{A}_0\mathrm{e}^{-\lambda t}\Rightarrow\lambda\mathrm{t}=\ln 8$

$\lambda\mathrm{t}=3\ln 2$

$\frac{\ln 2}{\lambda}=\frac{\mathrm{t}}{3}=\frac{30}{3}=10$ years

Q10

$N = N_0 e^{-\lambda t}$

$N_d = N_0 - N$

$N_d = N_0\left(1 - e^{-\lambda t}\right)$

$\frac{N_d}{N_0} = f = 1 - e^{-\lambda t}$

$\frac{df}{dt} = \lambda e^{-\lambda t}$

Q11

$T_m = 30$ ms

$C = 200\mu F$

$\frac{q}{N} = \frac{Q_0 e^{-t/RC}}{N_0 e^{-\lambda t}} = \frac{Q_0}{N_0} e^{\left\{\left(\lambda - \frac{1}{RC}\right)\right.}$

Since q/N is constant hence

$\lambda = \frac{1}{RC}$

$R = \frac{1}{\lambda C} = \frac{T_m}{C} = \frac{30 \times 10^{-3}}{200 \times 10^{-6}} = 150\Omega$

Q12

(A) True, atom of each element emits characteristic spectrum.

(B) True, according to Bohr's postulates

$mvr = \frac{nh}{2\pi}$ and hence electron resides into orbits of specific radius called stationary orbits.

(C) False, density of nucleus is constant

(D) False, A free neutron is unstable decays into proton and electron and antineutrino.

(E) True unstable nucleus show radioactivity.